

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	14-07-2026
Project Name	Customer Registry

Technical Architecture:

The Customer Complaint Management System is a MERN Stack based web application developed to manage customer complaints efficiently.

The system consists of three user roles: Customer, Admin, and Agent.

Customers can register complaints and track their complaint status. Admins can manage users, assign complaints to agents, and monitor the complaint resolution process. Agents can view assigned complaints and update complaint statuses.

The frontend is developed using React.js and Bootstrap. The backend is developed using Node.js and Express.js. MongoDB is used as the database for storing user and complaint information. JWT authentication is implemented to provide secure access and role-based authorization.

Architecture Diagram:

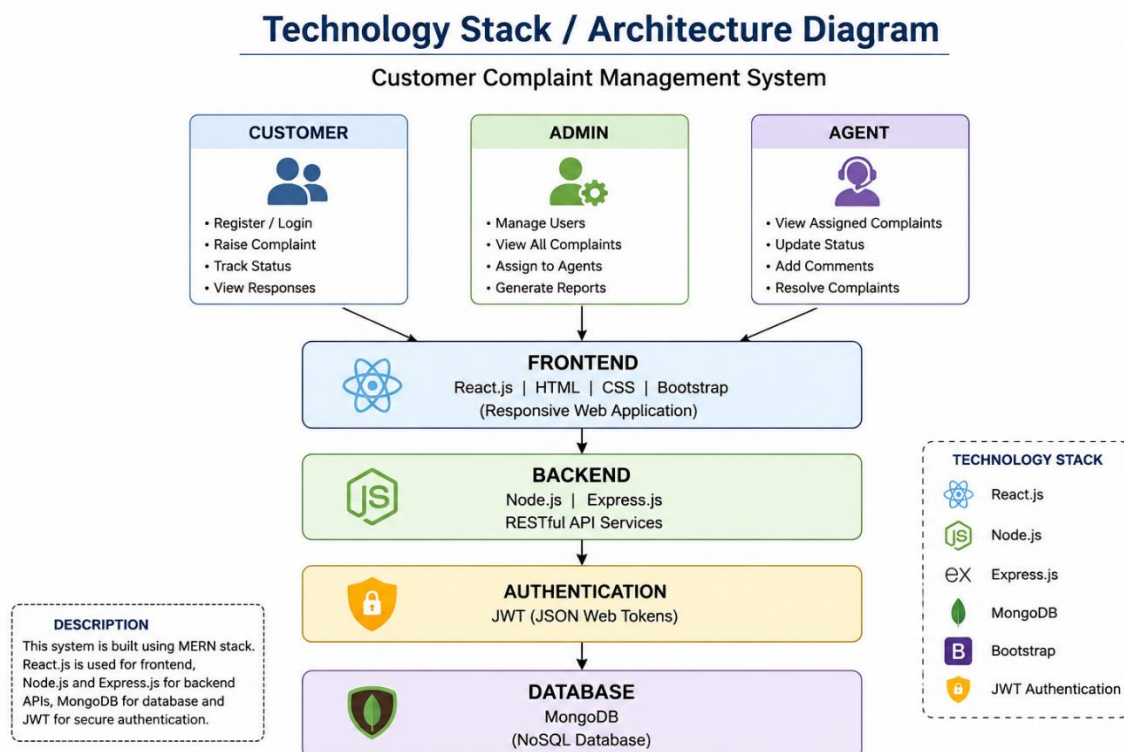


Table 1 : Components & Technologies

S. No	Component	Description	Technology
1.	User Interface	Customer, Admin and Agent Dashboards	React.js, Bootstrap
2.	Application Logic-1	User Registration and Login	Node.js, Express.js
3.	Application Logic-2	Complaint Management	Node.js, Express.js
4.	Application Logic-3	Complaint Assignment and Status Update	Node.js, Express.js
5.	Database	Stores User and Complaint Data	MongoDB
6.	Cloud Database	Database Hosting	MongoDB Atlas / Local MongoDB
7.	File Storage	Stores Project Files	Local Storage
8.	External API-1	HTTP Requests	Axios
9.	External API-2	REST APIs	Express APIs
10.	Machine Learning Model	Not Applicable	N/A
11.	Infrastructure	Application Execution Environment	VS Code, Node.js

Table – 2 : Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open Source Frameworks	Frontend and Backend Development Frameworks	React.js, Express.js
2.	Security Implementations	Authentication and Route Protection	JWT, bcryptjs, Middleware
3.	Scalable Architecture	Modular REST API Architecture	Node.js, Express.js
4.	Availability	Accessible through Web Browser	React.js
5.	Performance	Fast Data Retrieval and API Response	MongoDB, Express.js